Hierarchical Templates and Fleet Management

> **CaC-ConfigMgr** - Centralized LogPoint Configuration Management
> *Configuration as Code for LogPoint SIEM*

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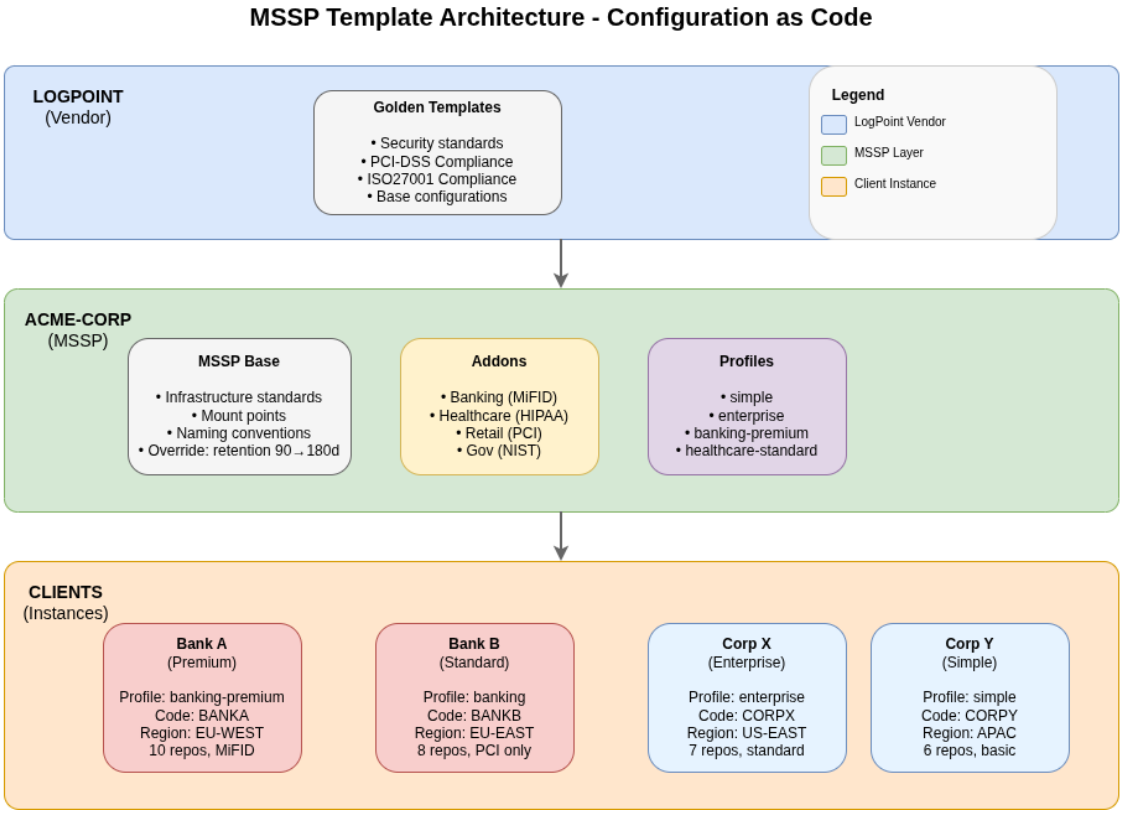
1. Context & Challenges

The Problem Before CaC

Before	After with CaC
50+ SIEMs managed manually	1 codebase = 50+ deployments
Frequent human errors	Automatic validation before deployment
No change traceability	Git = complete history
Client onboarding = 2 weeks	Onboarding = 1 YAML file
Mass update = nightmare	1 template change → all clients

Our Template MSSP Architecture



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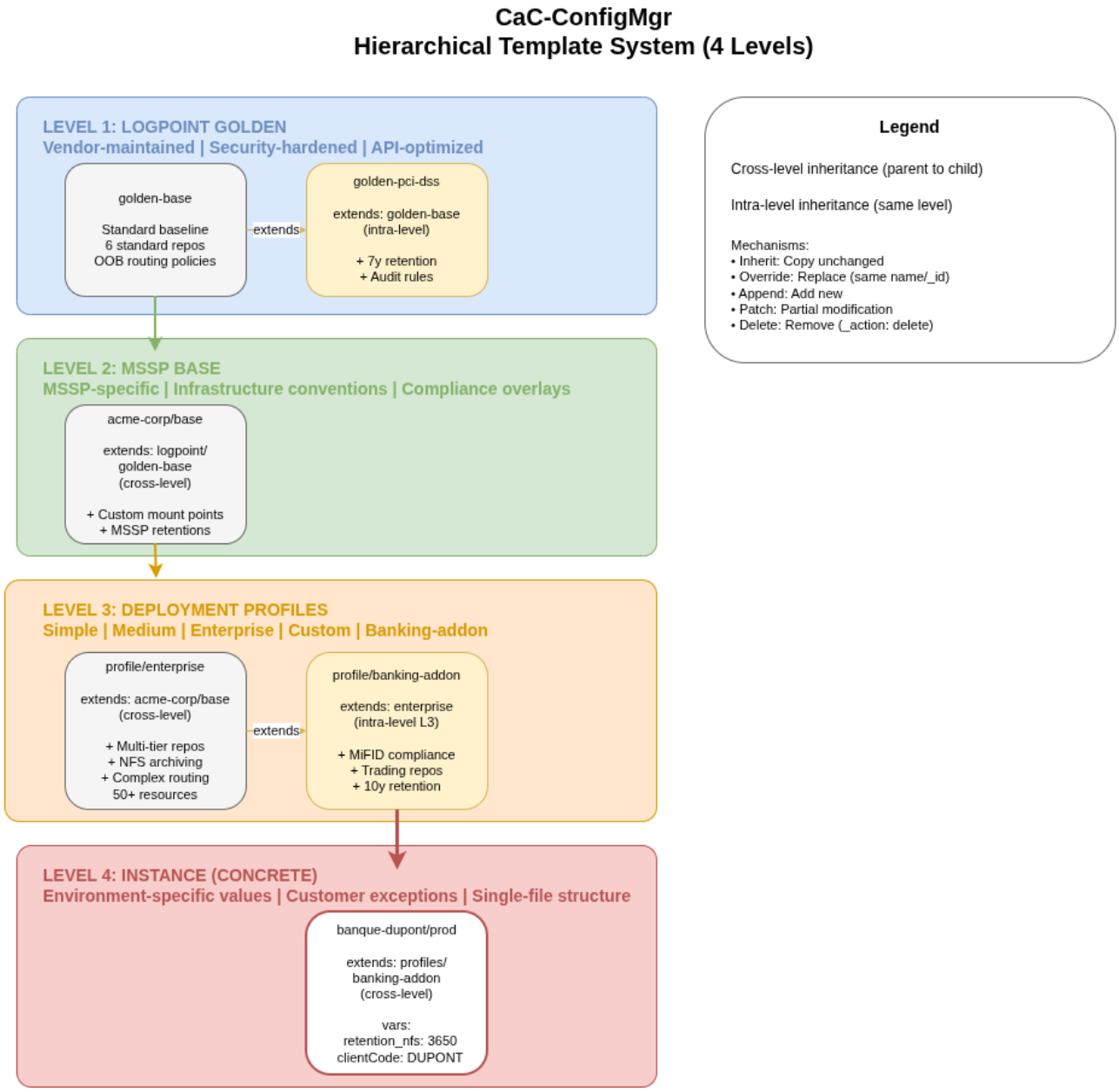
2. Key Concepts

2.1 Template Hierarchy (6 Levels)

Multi-level inheritance system enabling reuse and progressive specialization of configurations.



2.2 Fleet Management

A **Fleet** = a group of LogPoint nodes (DataNodes, SearchHeads, AIOS) sharing the same configuration.

```yaml
apiVersion: cac-configmgr.io/v1
kind: Fleet
metadata:
  name: bank-a
spec:
  managementMode: director
  director:
    poolUuid: pool-bank-a
    apiHost: https://director.logpoint.com
    credentialsRef: env://DIRECTOR_TOKEN
  nodes:
    aios: []
    dataNodes:
    # Production cluster with 2 DataNodes
```

```yaml
      - name: dn-bank-a-01
        logpointId: lp-bank-a-p1
        tags:
        - cluster: production
        - env: prod
        - region: eu-west-1
      - name: dn-bank-a-02
        logpointId: lp-bank-a-p2
        tags:
        - cluster: production
        - env: prod
        - region: eu-west-1
      searchHeads:
      # Search Head sees the production cluster
      - name: sh-bank-a-01
        logpointId: lp-bank-a-s1
        tags:
        - cluster: frontend
        - env: prod
        - sh-for: production
```

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3. Template Architecture

3.1 Vertical vs Horizontal Inheritance

Type	Direction	Description	Example
Cross-Level (Vertical)	N-1 → N	Standard parent→child inheritance	`mssp-base` extends `golden-base`
Intra-Level (Horizontal)	N → N	Same level, specialized add-ons	`banking-addon` extends `enterprise`

3.2 Concrete simplify example

Deep Merge and List Merging with _id

Parent Template
golden-base/repos.yaml

```
repos:
  - name: repo-secu
    hiddenrepopath:
      - _id: fast-tier
        path: /opt/immune/storage
        retention: 365
      - _id: warm-tier
        path: /opt/immune/storage-warm
        retention: 90
```

deep_merge()

Child Template
acme-enterprise/repos.yaml

```
repos:
  - name: repo-secu
    hiddenrepopath:
      - _id: fast-tier
        retention: 7          <-- Override
      - _id: warm-tier
        retention: 30         <-- Override
      - _id: nfs-tier        <-- Append (new)
        path: /opt/immune/storage-nfs
        retention: 1095
```

Final Resolution
(API Payload)

```
{
  "repos": [{
    "name": "repo-secu",
    "hiddenrepopath": [
      {"path": "/opt/immune/storage", "retention": 7},
      {"path": "/opt/immune/storage-warm", "retention": 30},
      {"path": "/opt/immune/storage-nfs", "retention": 1095}
    ]
  }]
}

Note: _id fields are filtered before API call
```

Ordering Directives

```
_after: _id      -- Insert after element
_before: _id     -- Insert before element
_position: N     -- Absolute position (1-based)
_first: true     -- Force first position
_last: true      -- Force last position

Example:
- _id: crit-powershell
  _after: crit-verbose
  repo: repo-secu
  key: CommandLine
```

Deletion (_action: delete)

```
To remove an inherited element:

routing_criteria:
  - _id: crit-obsolete
    _action: delete
```

Summary from our demo data:

Level	Template	What it brings
1	golden-base	• 6 standard repos • Basic routing policies • Retention: 90 days
2 (addon)	golden-pci-dss	• +1 PCI-audit repo • PCI retention: 7 years (2555d) • PCI processing policies
3	acme-base (MSSP)	• Override retention: 90 → 180d • Adds mount_warm, mount_cold • Archive tiering
4 (addon)	acme-banking-addon	• +1 trading repo • Banking retention: 10 years • MiFID compliance
5 (profile)	acme-banking-premium	• MiFID enrichment • Trading logs normalization • 4-tier storage (fast/warm/cold/nfs)

Level	Template	What it brings
6 (instance)	bank-a-prod	• client_code: "BANKA" • region: "EU-WEST" • Specific repo-secu override

3.3 Inheritance Mechanisms

Mechanism	Description	Use Case
Inherit	Copy all resources from parent unchanged	Standard compliance baselines
Override	Replace resource with same identifier	Change retention, modify paths
Append	Add new resources not in parent	Additional repos, custom policies
Patch	Partial modification of parent resource	Add a routing criteria, extend list
Delete	Remove resource from parent (explicit)	Disable default repo for specific case

3.4 Resource Identification & Merging

Top-level resources are identified by their **kind + name** tuple:

```
kind: RepoConfig
name: repo-secu  # ← Unique identifier for inheritance
```

For **lists** (e.g., `hiddenrepopath`, `routing_criteria`), elements are matched using **_id**:

```
repos:
  - name: repo-secu
    hiddenrepopath:
      - _id: fast-tier        # ← Template ID (internal, not sent to API)
        path: /opt/immune/storage
        retention: 30

      - _id: warm-tier        # ← Template ID for second element
        path: /opt/immune/storage-warm
        retention: 90
```

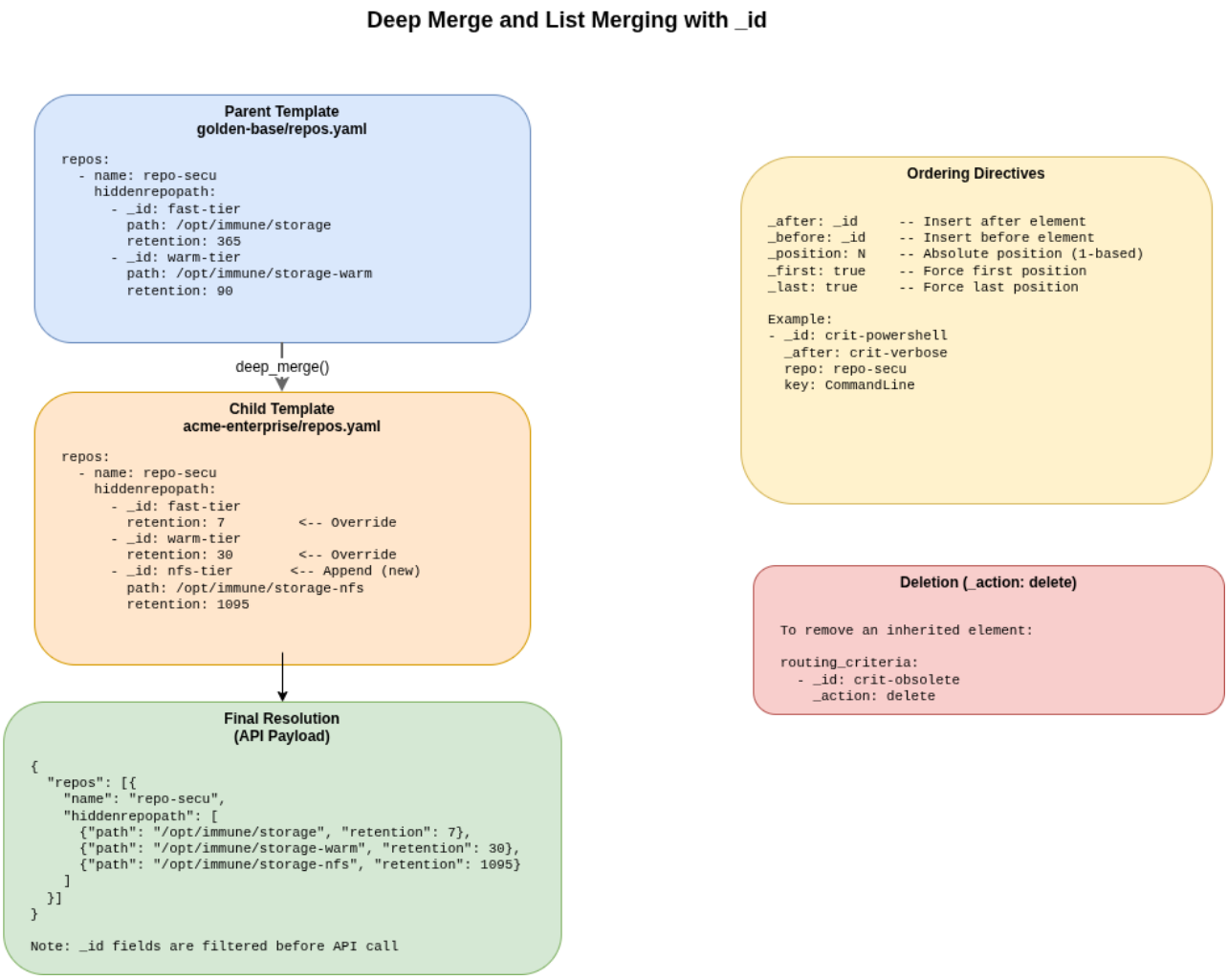
Matching Rules:

Scenario	Result
Same `_id` in parent and child	**Merge**: Child fields override parent, undefined fields inherited
`_id` only in parent	**Inherit**: Element kept as-is
`_id` only in child	**Append**: New element added to list

Scenario	Result
`_action: delete` on existing `_id`	**Remove**: Element removed from list

> **Important**: `_id` and `_action` fields are **internal to CaC-ConfigMgr** and are **filtered out** before sending to LogPoint API.

3.5 Merge Mechanisms



3.6 Ordering Directives

When merging lists (especially for **normalization policies**), element order matters critically. If a generic package is placed before a specific one, the log will be matched by the generic parser and the specific one will never be reached.

Directive	Description	Example
`_first: true`	Insert at the beginning of the list	Insert specific parsers first
`_last: true`	Insert at the end of the list	Append generic fallback parsers
`_position: N`	Insert at absolute position N	Fixed position requirement
`_after: element_id`	Insert after a specific element	Add after existing package
`_before: element_id`	Insert before a specific element	Add before existing package

Example: Normalization Policy Ordering

```
# Level 1: Parent template defines base normalization
normalizationPolicies:
  - name: np-auto
    normalization_packages:
      - _id: pkg-autoparser
        name: AutoParser  # ← Generic parser (catches everything)

# Level 3: MSSP adds specific Windows parsing
normalizationPolicies:
  - name: np-windows
    normalization_packages:
      - _id: pkg-windows
        name: Windows
        _before: pkg-autoparser   # ← MUST be before AutoParser!
      - _id: pkg-winsec
        name: WinSecurity
        _before: pkg-autoparser   # ← MUST be before AutoParser!
```

Why Order Matters:

```
× WRONG ORDER (generic first):
1. AutoParser (generic) → Matches ALL logs, including Windows
2. Windows (specific)   → NEVER REACHED!
3. WinSecurity          → NEVER REACHED!

✅ CORRECT ORDER (specific first):
1. Windows (specific)   ← Windows logs matched here
2. WinSecurity          ← Windows security events matched here
3. AutoParser (generic) ← Everything else falls back here
```

Example: Banking-specific SWIFT Parser

```
# Level 4: Banking addon adds SWIFT financial messaging
normalizationPolicies:
  - name: np-banking
    normalization_packages:
      - _id: pkg-swift
        name: SWIFT
        _first: true   # ← SWIFT messages MUST be parsed first
      - _id: pkg-sepa
        name: SEPA
        _first: true   # ← SEPA messages MUST be parsed first
```

Final Order After Merge:

```
np-banking normalization_packages:
1. SWIFT    ← _first: true
2. SEPA     ← _first: true
3. Windows  ← from parent
4. WinSecurity ← from parent
5. AutoParser  ← from parent
```

Precedence Rules

When multiple ordering directives conflict, the following precedence applies:

1. `_action: delete` - Remove element (highest priority)
2. `_position: N` - Absolute positioning
3. `_first: true` / `_last: true` - Relative positioning
4. `_after` / `_before` - Relative to other elements

> **Note**: Ordering directives are **internal to CaC-ConfigMgr** and are filtered out before API calls, like `_id` fields.

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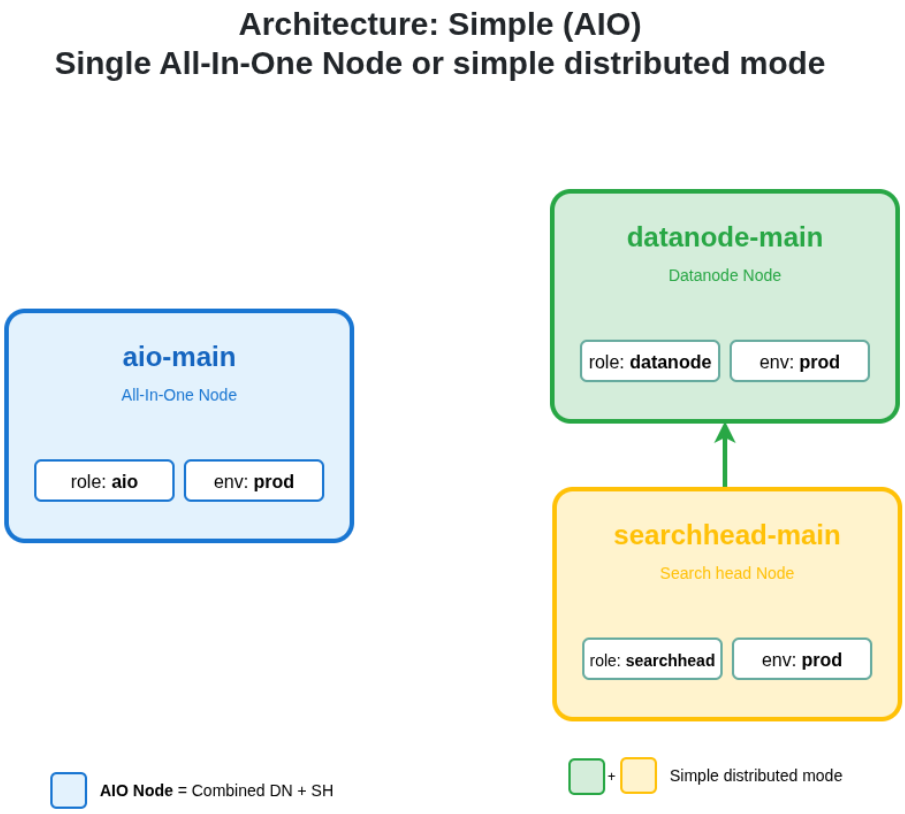
4. Fleet Management

4.1 Node Types

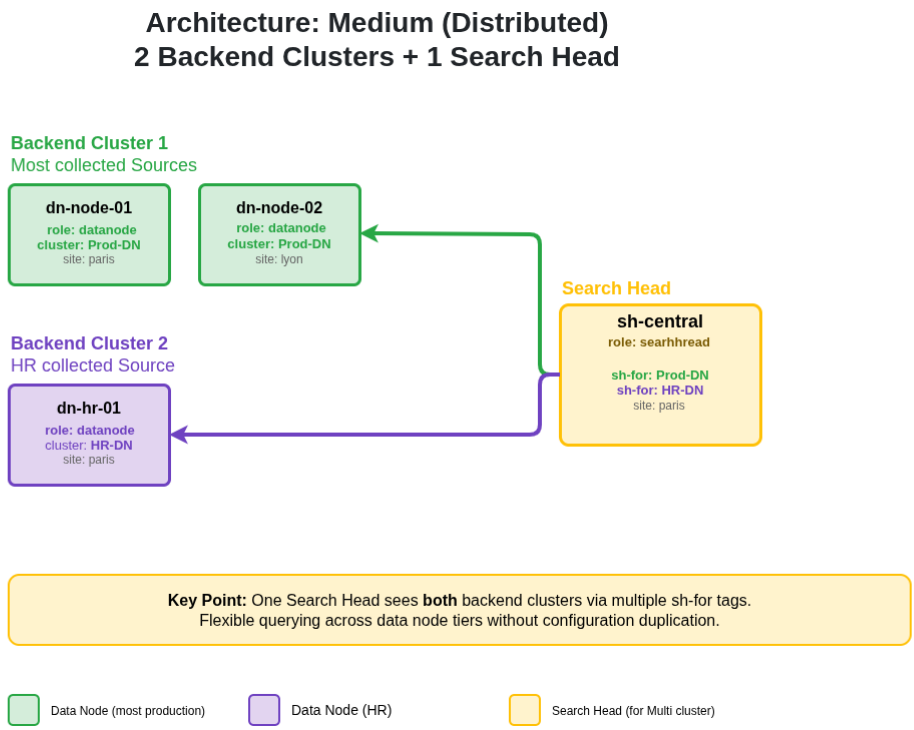
Type	Role	Example tags
DataNode	Storage and indexing	`cluster: production`, `env: prod`, `region: eu-west-1`
SearchHead	User interface	`cluster: frontend`, `env: prod`, `sh-for: production`
AIOS	All-in-One (combined DN+SH)	`cluster: dr-site`, `env: prod`

4.2 Supported Topologies

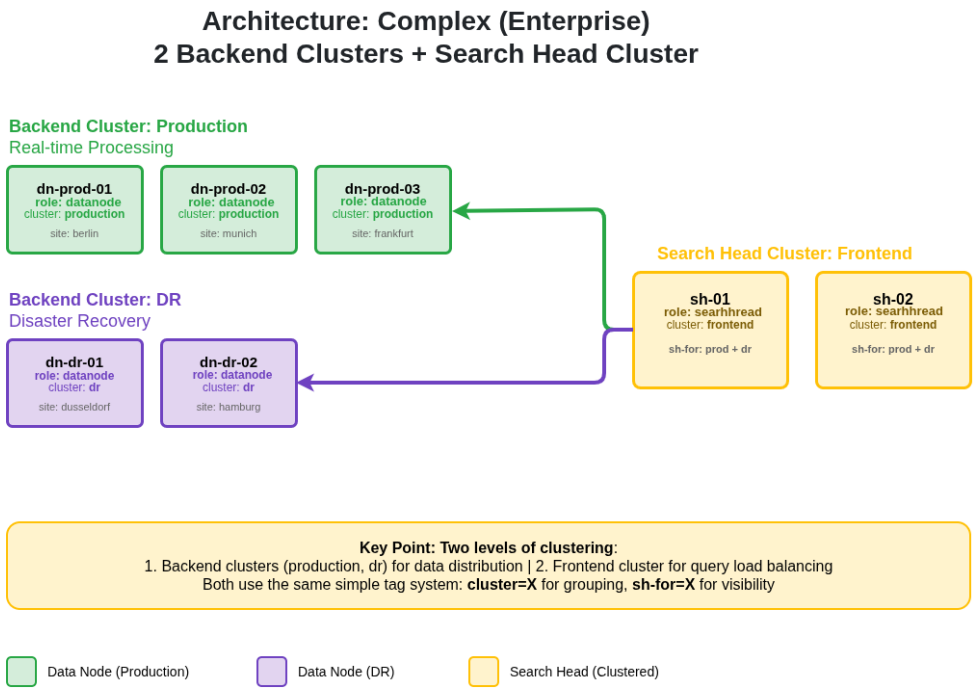
Standalone Mode (All-in-One or simple distributed mode)



Medium Simple Distributed Mode



Distributed with Clustering



4.3 Cluster Prerequisites

> ⚠ **Important**: To push identical configurations to a cluster of nodes, the following requirements must be met:

Requirement	Description
SIEM Version	All nodes in the cluster must run the **same LogPoint version** (e.g., 7.6, 7.7)
Plugin List	All nodes must have the **identical list of plugins** installed
Plugin Versions	Each plugin must be the **same version** across all nodes

Why this matters:

- Version mismatch can cause configuration deployment failures
- Missing plugins on one node will result in invalid configuration states
- The CaC tool validates these prerequisites before deployment

4.4 Tag System

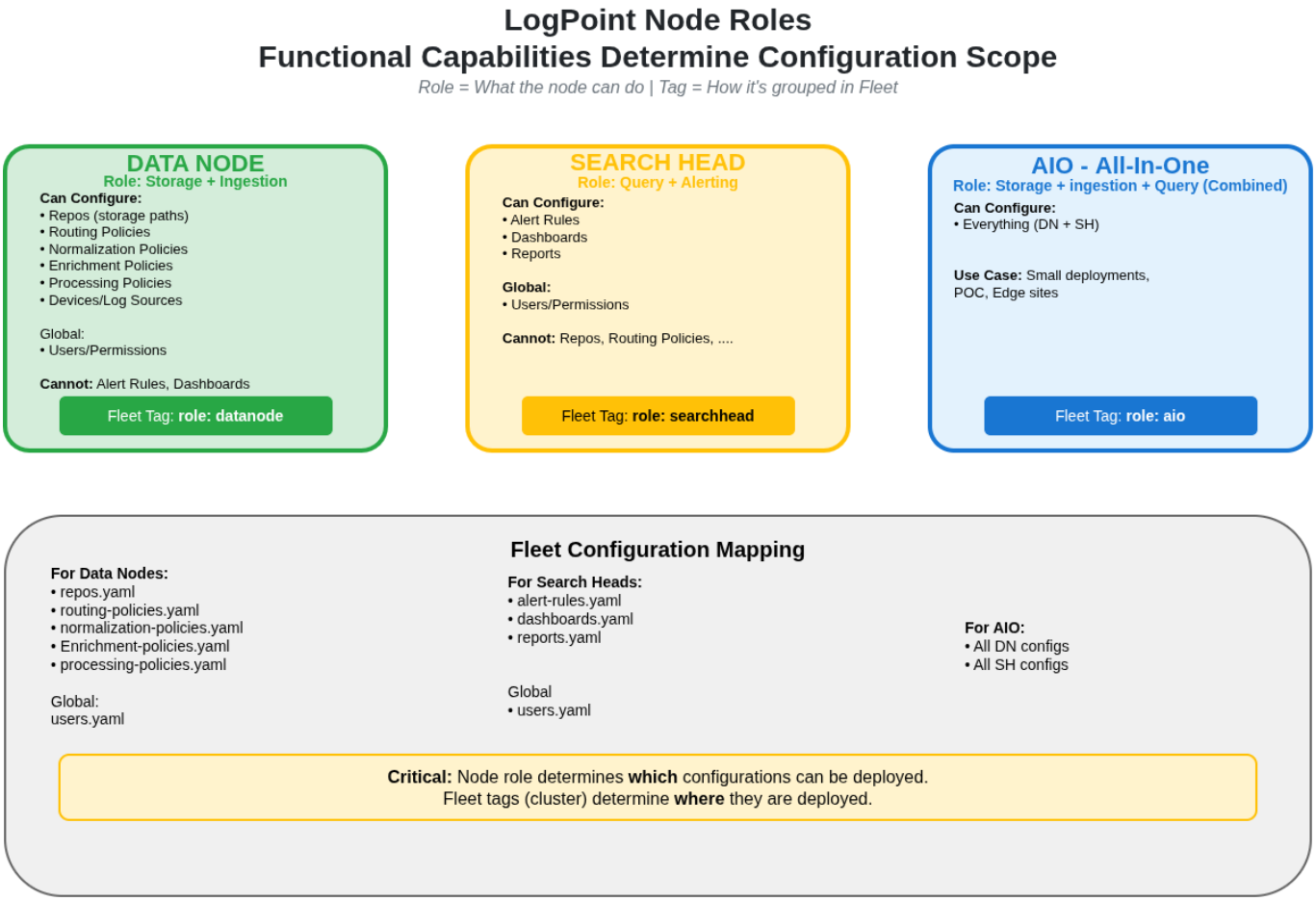
Tags define relationships between nodes:

```
# Cluster Membership
dn-prod-01: tags: [cluster: production]
dn-prod-02: tags: [cluster: production]  # Same cluster

# SH to DN Relationships
sh-prod-01:
  tags:
    - cluster: sh-frontend
    - sh-for: production   # ← See all DNs with cluster: production
```

Tag on SH	Meaning
`sh-for: production`	Sees all DNs with `cluster: production`
`sh-for: legacy`	Sees all DNs with `site: legacy`
No `sh-for` tag	Sees nothing (or everything if admin)

4.5 Node Roles



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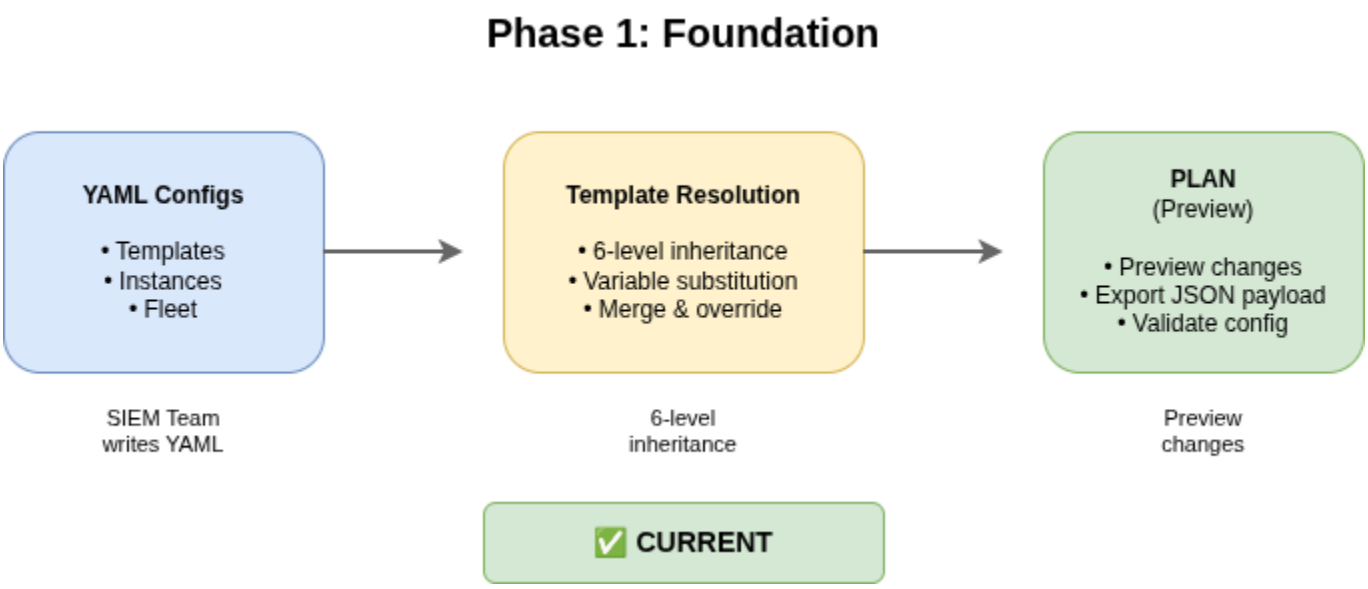
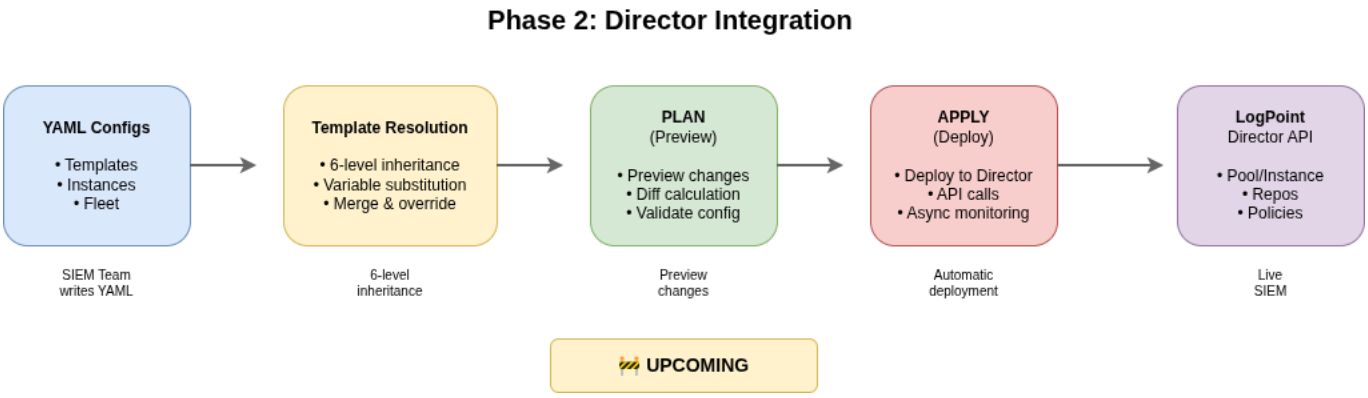
5. Demo - Operating Mode

5.1 "Desired State" Workflow

Phase 1 (Current): Foundation



Phase 2 (Upcoming): Director Integration



5.2 Main Commands (Phase 1)

```
# 1. VALIDATE - Check syntax and consistency
cac-configmgr validate demo-configs/
# → Check: YAML syntax, references, schemas

# 2. PLAN - Preview changes (generates desired state)
cac-configmgr plan \
  --fleet demo-configs/instances/banks/bank-a/prod/fleet.yaml \
  --topology demo-configs/instances/banks/bank-a/prod/instance.yaml \
  --templates-dir demo-configs/templates \
  --export-dir demo-configs/.desired-state/
# → Displays: resolved resources, variables, inheritance chain
# → Exports: JSON payload ready for API (Phase 2)
```

Phase 2 Commands (Upcoming):

```
# 3. APPLY - Deploy to Director/SIEM
cac-configmgr apply \
  --fleet demo-configs/instances/banks/bank-a/prod/fleet.yaml \
```

```
  --topology demo-configs/instances/banks/bank-a/prod/instance.yaml \
  --auto-approve
# → Deploys: Configuration to LogPoint via Director API
```

5.3 Client Configuration Example

Below is a minimal instance configuration. See the Appendix for the complete hierarchy (Levels 1-6) and fleet configuration with clustered DataNodes.

```yaml
# instances/banks/bank-a/prod/instance.yaml
apiVersion: cac-configmgr.io/v1
kind: TopologyInstance
metadata:
  name: bank-a-prod
  extends: mssp/acme-corp/profiles/banking-premium  # ← Inherits premium
profile
  fleetRef: ./fleet.yaml                             # ← References fleet

spec:
  vars:
    client_code: BANKA        # ← Client-specific variable
    region: EU-WEST           # ← Client-specific variable

  # Specific overrides (optional)
  repos:
    - name: repo-secu
      hiddenrepopath:
        - _id: nfs-tier
          retention: 3650     # ← Override: 10 years for this client
```

---

6. Key Benefits

6.1 DRY Principle (Don't Repeat Yourself)

```
× Before: 50 clients × 100 config lines = 5000 lines to maintain
✅ After: 6 templates + 50 instance files = ~800 lines only
```

6.2 Simplified Mass Updates

```
# Modify retention_default in acme-base
# → Impact: ALL ACME clients are automatically updated
```

6.3 Fast Configuration Onboarding

```
# New client configuration = 1 YAML file
# Configuration time: from week(s) to hour(s)
# (Machine provisioning time remains unchanged)
```

6.4 Guaranteed Compliance

```
PCI-DSS Addon → all banking clients inherit
ISO27001 Addon → all sensitive clients inherit
```

6.5 Audit and Traceability

```
Git log = complete change history
Who changed what when = immediately available
```

🎤 Key Messages to Communicate

1. **"One codebase to manage 1 or 100 SIEMs with the same effort"**
2. **"6-level inheritance allows factoring up to 90% of the configuration (based on client complexity)"**
3. **"Validation before deployment = zero errors in production"**
4. **"Client onboarding in hours, not weeks"**
5. **"Compliance by construction, not by verification"**

📎 Available Diagrams

All diagrams are located in `docs/Presentations/TemplateHierarchy-Fleet/images/`:

Diagram	File	Description
Node Roles	`01-node-roles.png`	DataNode, SearchHead, AIO capabilities and fleet tags
Simple Architecture	`02-simple-arch.png`	Single AIO or simple distributed mode
MSSP Architecture	`03-mssp-architecture.png`	3-tier MSSP architecture (Vendor → MSSP → Clients)
Medium Distributed	`03-medium-dist.png`	2 backend clusters + 1 Search Head
Complex Enterprise	`04-complex-ent.png`	Production cluster + DR + SH cluster

Diagram	File	Description
Workflow Phase 1	`04-workflow-phase1.png`	Phase 1: Foundation workflow (YAML → Resolution → Plan)
Workflow Phase 2	`04-workflow-phase2.png`	Phase 2: Director Integration workflow (+ Apply)
Deployment Flow	`05-deployment-flow.png`	Complete deployment pipeline (Resolution → Substitution → Filtering → API)
4-Level Overview	`05-4level-overview.png`	Hierarchical template system overview
Merge Mechanisms	`06-merge-mechanisms.png`	Deep merge with `_id` matching
Complete Example	`07-complete-example.png`	Full repo inheritance chain

📚 References

Document	Description
`specs/20-TEMPLATE-HIERARCHY.md`	Complete template hierarchy specification
`specs/10-INVENTORY-FLEET.md`	Fleet Inventory specification
`docs/DEMO-OPERATING-MODE.md`	Complete demo guide
`examples/`	Configuration examples

Appendix: Complete Bank A Example

This section provides a complete, concrete example of the Bank A production configuration, showing all inheritance levels from Golden Base to the final API payload.

Inheritance Chain Overview

```
bank-a-prod (Level 6)
└── extends: acme-banking-premium (Level 5)
    └── extends: acme-banking-addon (Level 4)
        └── extends: acme-base (Level 3)
            └── extends: golden-pci-dss (Level 2)
                └── extends: golden-base (Level 1)
```

Level 1: LogPoint Golden Base

File: `templates/logpoint/golden-base/repos.yaml`

```yaml
apiVersion: cac-configmgr.io/v1
kind: ConfigTemplate
metadata:
  name: golden-base
  version: 2.0.0
  provider: logpoint
spec:
  repos:
  - name: repo-default
    hiddenrepopath:
    - _id: primary
      path: '{{mount_point}}'
      retention: '{{retention_default}}'
  - name: repo-secu
    hiddenrepopath:
    - _id: primary
      path: '{{mount_point}}'
      retention: '{{retention_sec}}'
  - name: repo-secu-verbose
    hiddenrepopath:
    - _id: primary
      path: '{{mount_point}}'
      retention: 30
  - name: repo-system
    hiddenrepopath:
    - _id: primary
      path: '{{mount_point}}'
      retention: 180
  - name: repo-system-verbose
    hiddenrepopath:
    - _id: primary
      path: '{{mount_point}}'
      retention: 30
  - name: repo-cloud
    hiddenrepopath:
    - _id: primary
      path: '{{mount_point}}'
      retention: 180
```

Variables: `templates/logpoint/golden-base/vars.yaml`

```yaml
apiVersion: cac-configmgr.io/v1
kind: ConfigTemplate
metadata:
  name: golden-base
spec:
  vars:
    mount_point: /opt/immune/storage
    retention_default: 90
    retention_sec: 365
```

Level 2: PCI-DSS Addon (Intra-level)

File: `templates/logpoint/golden-pci-dss/repos.yaml`

```yaml
apiVersion: cac-configmgr.io/v1
kind: ConfigTemplate
metadata:
  name: golden-pci-dss
  extends: logpoint/golden-base
  version: 1.0.0
  provider: logpoint
spec:
  repos:
  - name: repo-pci-audit
    hiddenrepopath:
    - _id: primary
      path: /opt/immune/storage-nfs
      retention: '{{retention_pci}}'
```

Variables: `templates/logpoint/golden-pci-dss/vars.yaml`

```yaml
apiVersion: cac-configmgr.io/v1
kind: ConfigTemplate
metadata:
  name: golden-pci-dss
  extends: logpoint/golden-base
spec:
  vars:
    retention_pci: 2555  # 7 years PCI requirement
```

Level 3: MSSP Base (ACME Corp)

File: `templates/mssp/acme-corp/base/repos.yaml`

```yaml
apiVersion: cac-configmgr.io/v1
kind: ConfigTemplate
metadata:
  name: acme-base
  extends: logpoint/golden-pci-dss
  version: 1.0.0
  provider: acme-mssp
spec:
  repos:
  - name: repo-secu
    hiddenrepopath:
```

```
      - _id: primary
        retention: 90  # Override from parent
      - _id: warm-tier
        path: '{{mount_warm}}'
        retention: 365
    - name: repo-archive
      hiddenrepopath:
      - _id: warm-tier
        path: '{{mount_warm}}'
        retention: 90
      - _id: cold-tier
        path: '{{mount_cold}}'
        retention: 1095
```

Variables: `templates/mssp/acme-corp/base/vars.yaml`

```
apiVersion: cac-configmgr.io/v1
kind: ConfigTemplate
metadata:
  name: acme-base
  extends: logpoint/golden-pci-dss
spec:
  vars:
    retention_default: 180  # Override: 90→180
    mount_warm: /opt/immune/storage-warm
    mount_cold: /opt/immune/storage-cold
```

Level 4: Banking Addon (Intra-level)

File: `templates/mssp/acme-corp/addons/banking/repos.yaml`

```
apiVersion: cac-configmgr.io/v1
kind: ConfigTemplate
metadata:
  name: acme-banking-addon
  extends: mssp/acme-corp/base
  version: 1.0.0
  provider: acme-mssp
spec:
  repos:
  - name: repo-trading
    hiddenrepopath:
    - _id: primary
      path: '{{mount_point}}'
      retention: 1
    - _id: warm-tier
      path: '{{mount_warm}}'
      retention: 7
```

```
    - _id: nfs-tier
      path: /opt/immune/storage-nfs
      retention: 2555
```

Variables: `templates/mssp/acme-corp/addons/banking/vars.yaml`

```yaml
apiVersion: cac-configmgr.io/v1
kind: ConfigTemplate
metadata:
  name: acme-banking-addon
  extends: mssp/acme-corp/base
spec:
  vars:
    retention_banking: 3650  # 10 years MiFID
```

Level 5: Banking Premium Profile

File: `templates/mssp/acme-corp/profiles/banking-premium/repos.yaml`

```yaml
apiVersion: cac-configmgr.io/v1
kind: ConfigTemplate
metadata:
  name: acme-banking-premium
  extends: mssp/acme-corp/addons/banking
  version: 1.0.0
  provider: acme-mssp
spec:
  repos:
  - name: repo-banking-archive
    hiddenrepopath:
    - _id: primary
      path: /opt/immune/storage-nfs
      retention: 3650
```

Variables: `templates/mssp/acme-corp/profiles/banking-premium/vars.yaml`

```yaml
apiVersion: cac-configmgr.io/v1
kind: ConfigTemplate
metadata:
  name: acme-banking-premium
  extends: mssp/acme-corp/addons/banking
spec:
  vars:
    compliance: mifid-banking
```

Level 6: Bank A Production Instance

File: `instances/banks/bank-a/prod/instance.yaml`

```yaml
apiVersion: cac-configmgr.io/v1
kind: TopologyInstance
metadata:
  name: bank-a-prod
  extends: mssp/acme-corp/profiles/banking-premium
  fleetRef: ./fleet.yaml
spec:
  vars:
    client_code: BANKA
    region: EU-WEST
  repos:
  - name: repo-secu
    hiddenrepopath:
    - _id: nfs-tier        # Adding new tier
      path: /opt/immune/storage-nfs
      retention: 3650      # 10 years for this client
```

Fleet Configuration

File: `instances/banks/bank-a/prod/fleet.yaml`

```yaml
apiVersion: cac-configmgr.io/v1
kind: Fleet
metadata:
  name: bank-a
spec:
  managementMode: director
  director:
    poolUuid: pool-bank-a
    apiHost: https://director.logpoint.com
    credentialsRef: env://DIRECTOR_TOKEN
  nodes:
    aios: []
    dataNodes:
    # Production cluster with 2 DataNodes
    - name: dn-bank-a-01
      logpointId: lp-bank-a-p1
      tags:
      - cluster: production
      - env: prod
      - region: eu-west-1
    - name: dn-bank-a-02
      logpointId: lp-bank-a-p2
      tags:
      - cluster: production
```

```
        - env: prod
        - region: eu-west-1
      searchHeads:
      # Search Head sees the production cluster
      - name: sh-bank-a-01
        logpointId: lp-bank-a-s1
        tags:
        - cluster: frontend
        - env: prod
        - sh-for: production
```

Final Merged Configuration

After template resolution and variable substitution, the following configuration is generated for Bank A:

Resolved Variables

```
mount_point: /opt/immune/storage
retention_default: 180          # Overridden by acme-base
retention_sec: 365              # From golden-base
retention_pci: 2555             # From golden-pci-dss
mount_warm: /opt/immune/storage-warm
mount_cold: /opt/immune/storage-cold
retention_banking: 3650
compliance: mifid-banking
client_code: BANKA
region: EU-WEST
```

Resolved Repos (10 total)

Repo	Path(s)	Retention	Source Level
repo-default	/opt/immune/storage	180	Level 3 (acme-base override)
repo-secu	/opt/immune/storage + warm + nfs	90, 365, 3650	L1+L3+L6 merge
repo-secu-verbose	/opt/immune/storage	30	Level 1
repo-system	/opt/immune/storage	180	Level 1
repo-system-verbose	/opt/immune/storage	30	Level 1
repo-cloud	/opt/immune/storage	180	Level 1
repo-pci-audit	/opt/immune/storage-nfs	2555	Level 2
repo-archive	warm + cold	90 + 1095	Level 3

Repo	Path(s)	Retention	Source Level
repo-trading	storage + warm + nfs	1 + 7 + 2555	Level 4
repo-banking-archive	/opt/immune/storage-nfs	3650	Level 5

Final API Payload (JSON) - Repos Only

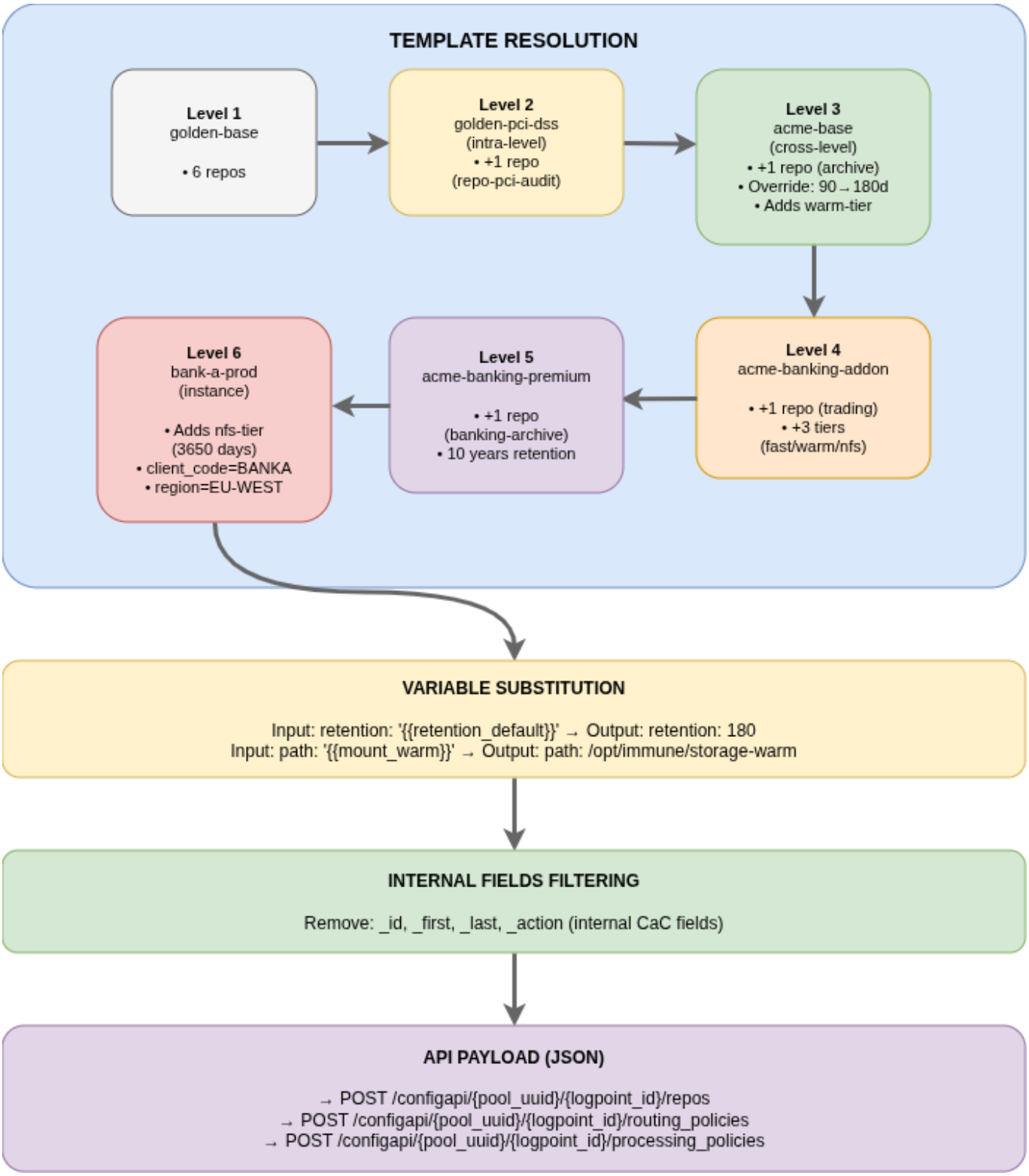
The following JSON payload (repos section) is sent to the LogPoint Director API after template resolution:

```
{
  "repos": [
    {
      "name": "repo-default",
      "hiddenrepopath": [
        {"path": "/opt/immune/storage", "retention": 180}
      ]
    },
    {
      "name": "repo-secu",
      "hiddenrepopath": [
        {"path": "/opt/immune/storage", "retention": 90},
        {"path": "/opt/immune/storage-warm", "retention": 365},
        {"path": "/opt/immune/storage-nfs", "retention": 3650}
      ]
    },
    {
      "name": "repo-secu-verbose",
      "hiddenrepopath": [
        {"path": "/opt/immune/storage", "retention": 30}
      ]
    },
    {
      "name": "repo-system",
      "hiddenrepopath": [
        {"path": "/opt/immune/storage", "retention": 180}
      ]
    },
    {
      "name": "repo-system-verbose",
      "hiddenrepopath": [
        {"path": "/opt/immune/storage", "retention": 30}
      ]
    },
    {
      "name": "repo-cloud",
      "hiddenrepopath": [
        {"path": "/opt/immune/storage", "retention": 180}
      ]
    },
    {
```

```
        "name": "repo-pci-audit",
        "hiddenrepopath": [
          {"path": "/opt/immune/storage-nfs", "retention": 2555}
        ]
      },
      {
        "name": "repo-archive",
        "hiddenrepopath": [
          {"path": "/opt/immune/storage-warm", "retention": 90},
          {"path": "/opt/immune/storage-cold", "retention": 1095}
        ]
      },
      {
        "name": "repo-trading",
        "hiddenrepopath": [
          {"path": "/opt/immune/storage", "retention": 1},
          {"path": "/opt/immune/storage-warm", "retention": 7},
          {"path": "/opt/immune/storage-nfs", "retention": 2555}
        ]
      },
      {
        "name": "repo-banking-archive",
        "hiddenrepopath": [
          {"path": "/opt/immune/storage-nfs", "retention": 3650}
        ]
      }
    ]
  }
```

> Note: Additional API calls are made for routing policies, processing policies, normalization policies, and enrichment policies.

Deployment Flow



Document prepared for CaC-ConfigMgr demonstration

Phase 1: Foundation (Template Resolution, Validation, Plan) | Phase 2: Director Integration (Apply)